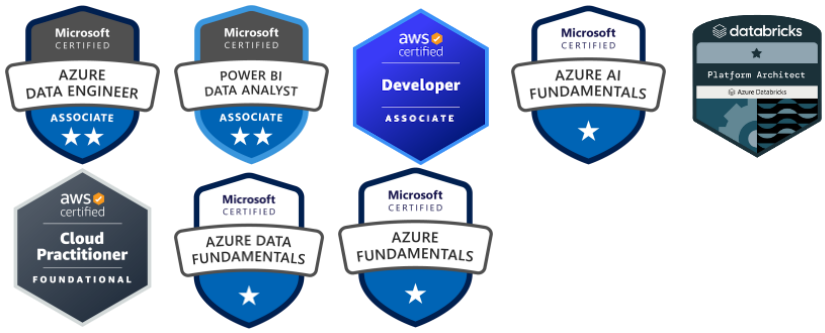
RAJ KUMAR MANALA
Senior Cloud Data Engineer
Email: rajkumarmanala23@gmail.com
Phone: +1(847)489-7388
Portfolio: https://rajkumarmanala.com
LinkedIn: https://www.linkedin.com/in/raj-kumar-manala-ab1b78134/



PROFESSIONAL SUMMARY:

- Over 6+ years of professional experience as a Senior Cloud Data Engineer specializing in architecting and delivering enterprise Lakehouse and data warehouse platforms, scalable ETL/ELT pipelines, and GenAI solutions across financial services, insurance, and IoT domains.
- Hands-on experience leading analytics platform modernization using Microsoft Fabric (OneLake, Lakehouse, Warehouse) with Medallion (Bronze/Silver/Gold) architecture to unify siloed financial datasets into a governed single source of truth.
- Strong expertise in dimensional modeling and data warehousing including Data Marts, OLAP/OLTP, Kimball methodology, star/snowflake schema design, SCD Type 1/2, and building facts/dimensions to standardize KPIs and enable self-service BI.
- Extensive experience building big data pipelines and transformations using Apache Spark (PySpark, Spark SQL, Structured Streaming) on Azure Databricks, Fabric Notebooks, HDInsight, and AWS EMR for multi-terabyte processing and optimization.
- Strong background in data integration and ingestion from diverse sources (SQL Server, MySQL, APIs, JSON, Kafka, Event Hubs) into Lakehouse architectures using ADF, Fabric Data Pipelines, Databricks, and metadata-driven frameworks.
- Experience designing and implementing data quality and validation frameworks including incremental loads, watermarking, idempotency, reconciliation checks, and automated error handling — consistently meeting 99.5%+ enterprise SLAs.
- Proficient in Agile/Scrum and end-to-end delivery across the SDLC including requirements, design, development, testing, CI/CD, performance tuning, deployments, monitoring/alerting, and production support in regulated financial environments.
- Advanced SQL and T-SQL skills across cloud data warehouses including Azure Synapse Analytics (Dedicated and Serverless Pools), Snowflake, AWS Redshift, Fabric Warehouse, and Azure SQL Database with focus on query optimization and performance tuning.
- Skilled in cloud data platform engineering across Azure and AWS, leveraging services such as ADF, ADLS Gen2, Databricks, Event Hubs, Synapse, Fabric, Purview, and AWS S3, Glue, EMR, Redshift, Lambda, Kinesis, and IAM for scalable architectures.
- Experienced in implementing cloud monitoring and observability using Azure Monitor, Log Analytics, Application Insights, Fabric Monitoring Hub, and AWS CloudWatch with proactive alerting and operational dashboards to ensure pipeline health.
- Strong knowledge of CI/CD and DevOps practices using Azure DevOps, GitHub Actions, Jenkins, Docker, and Kubernetes (AKS) including automated testing, version control, ARM template parameterization, and blue-green deployment strategies.
- Proven ability to build secure, governed data environments using Microsoft Purview (catalog, lineage, sensitive data classification), Entra ID RBAC, and Key Vault to protect regulated financial data (PII, PCI, SOX) and improve audit readiness.
- Proven experience supporting both batch and near real-time processing patterns to enable operational visibility and time-sensitive analytics across card transactions, dispute processing, fraud signals, and payment event tracking.
- Experienced in event streaming and real-time integration architectures using Azure Event Hubs, Fabric Eventstreams, Azure Stream Analytics, Kafka, and AWS Kinesis to support scalable ingestion and near real-time analytics pipelines.
- Strong communication and cross-functional collaboration skills, partnering with data science, compliance, business analytics, and engineering teams to translate complex requirements into scalable data solutions and measurable business outcomes.
- Experience delivering production-grade GenAI solutions using Azure OpenAI (GPT-5.2) + Azure AI Search to build RAG pipelines grounded in internal knowledge bases, reducing manual lookup effort and improving compliance and operational efficiency.
- Skilled in prompt engineering and structured output design (schema-valid JSON, reusable templates, few-shot patterns) to improve LLM reliability for document classification, dispute summarization, case routing, and compliance tagging workflows.
- Strong understanding of Responsible AI and enterprise security practices including Azure AI Content Safety guardrails, PII masking/redaction using Presidio, prompt-injection defenses, and audit logging aligned to enterprise risk standards.
- Experience establishing MLOps frameworks using Azure Machine Learning and MLflow for experiment tracking, model versioning, deployment governance, staged promotion pipelines, and drift/quality monitoring to ensure reproducible ML lifecycle management.

EDUCATION:

- Master of Science in Computer and Information Science — Aug 2023 – Dec 2025
  Concordia University Chicago, River Forest, IL, USA

- Bachelors in Electronics and Communication Engineering — July 2015 - Jan 2020
  Swami Vivekananda Institute of Technology, Hyderabad, India

TECHNICAL SKILLS:

Cloud Platforms	**Azure:** Azure Data Factory, ADLS Gen2, Azure Blob Storage, Azure Synapse Analytics, Azure Databricks, Azure HDInsight, Azure Event Hubs, Azure Stream Analytics, Azure Service Bus, Azure IoT Hub, Azure SQL Database, Azure Container Apps, AKS, Azure Key Vault, Azure Monitor, Azure Log Analytics, Application Insights, Entra ID, Microsoft Purview, ARM Templates, Microsoft Fabric **Databricks:** Azure Databricks, Delta Lake, Structured Streaming, CI/CD Integration, Git Integration **AWS:** S3, Glue, EMR, Lambda, Redshift, Athena, Kinesis, RDS (PostgreSQL), EC2, SNS, IAM, VPC, CloudWatch
Microsoft Fabric	OneLake, Lakehouse, Warehouse, Data Pipelines, Dataflows Gen2, Spark Notebooks, Event streams, Real-Time Analytics, Power BI Semantic Models, Deployment Pipelines, Workspace Management, Fabric Monitoring Hub, End-to-End Analytics
AI / GenAI / ML	Azure OpenAI Service (GPT-4, Chat, Embeddings), Azure AI Search, Azure AI Foundry, Prompt Flow, Azure AI Content Safety, RAG (Retrieval-Augmented Generation), Prompt Engineering, Azure Machine Learning, MLflow, Responsible AI, Presidio (PII Detection/Masking)
Programming Languages	Python, PySpark, Spark SQL, SQL, T-SQL, Scala, Java, Bash
Data Processing / Compute	Apache Spark (Spark SQL, Structured Streaming, Spark Streaming), Apache Kafka, Hive, HBase, Sqoop, Pig, Hadoop (HDFS)
Data Warehousing / Storage	Snowflake, Azure Synapse Analytics (Dedicated & Serverless SQL Pools), AWS Redshift, Delta Lake, Parquet
Visualization / Reporting	Power BI (Dashboards, Semantic Models, Row-Level Security, Scheduled Refresh), Tableau
Containerization & DevOps	Git, GitHub Actions, Jenkins, Azure DevOps, CI/CD, Docker, Kubernetes (AKS), ARM Templates
Databases	SQL Server, MySQL, PostgreSQL (AWS RDS), HBase
Collaboration & Tools	JIRA, Confluence, Azure DevOps Boards, Agile/Scrum
IDE's	VS Code, IntelliJ, Eclipse, Jupyter Notebook, PyCharm

PROFESSIONAL EXPERIENCE:

Client: Discover Financial **February 2025 - Present**
Role: Sr Cloud Data Engineer **Chicago, IL, USA**

Responsibilities:

- Led modernization of Discover's enterprise analytics platform by implementing a Microsoft Fabric Lakehouse on OneLake with Medallion (Bronze/Silver/Gold) architecture, unifying siloed financial datasets — card transactions, dispute cases, customer profiles, credit applications, and payment events — into a governed single source of truth for BI, analytics, and ML.
- Designed fault-tolerant, metadata-driven ingestion pipelines using Azure Data Factory and Fabric Data Pipelines with incremental loads, watermarking, idempotency, retry policies, and multi-stage validation — ingesting from 15+ source systems while maintaining 99.7%+ SLA compliance across batch and near real-time pipelines.
- Developed PySpark and Spark SQL transformations in Azure Databricks and Fabric Notebooks to cleanse, deduplicate, and enrich high-volume financial datasets — reducing processing time by ~40% through Delta Lake optimization, Z-order indexing, partition pruning, broadcast joins, and incremental MERGE INTO patterns.
- Modeled curated Gold-layer datasets in Fabric Warehouse using T-SQL with star-schema design (facts, dimensions, SCD Type 1/2), enabling consistent KPI definitions across card services, disputes, and banking domains — with governed Power BI semantic models, row-level security, and dataset certification.
- Implemented near real-time event processing using Azure Event Hubs and Fabric Eventstreams to capture card authorization signals, dispute triggers, payment events, and fraud alerts — reducing operational reporting latency from hours to under 5 minutes for time-sensitive dispute and fraud scenarios.
- Built Databricks-based pipelines processing structured and semi-structured financial data (JSON, Parquet, Delta) into curated analytical datasets — performing window functions, cross-account reconciliations, and business rule validations powering credit risk models, dispute trend analysis, and cardholder behavior segmentation.

- Delivered a production-grade RAG solution using Azure OpenAI (GPT-4) + Azure AI Search grounded in Discover's internal policies, dispute resolution guidelines, and compliance procedures — reducing manual document lookup effort by ~60% and improving response consistency for compliance and support teams.
- Built robust prompt engineering standards including few-shot libraries, reusable templates, and structured JSON output schemas for document classification, dispute summarization, and case routing — improving automation reliability from ~72% to 94% accuracy on production workloads.
- Designed agentic AI workflows using Azure AI Foundry and Prompt Flow to orchestrate multi-step automation (retrieve evidence → validate → generate → escalate) for dispute review and compliance verification — reducing case turnaround time by ~45% while maintaining human-in-the-loop controls for exceptions.
- Implemented Responsible AI controls including Azure AI Content Safety guardrails, PII masking/redaction using Presidio, prompt-injection defenses with input sanitization, and comprehensive audit logging — aligning all AI deployments with Discover's enterprise risk, compliance, and data privacy requirements.
- Established an MLOps framework using Azure Machine Learning + MLflow for experiment tracking, model versioning, automated validation gates, staged deployment pipelines, and continuous drift monitoring — ensuring reproducibility and operational stability for dispute classification, fraud scoring, and credit risk models.
- Deployed containerized Python microservices (REST APIs) on Azure Container Apps and AKS for ML inference and RAG endpoints — with GitHub Actions CI/CD automating testing, vulnerability scanning, Docker builds, and blue-green deployments with automated rollback across environments.
- Developed Power BI dashboards connected to Fabric Warehouse semantic models and Databricks SQL endpoints, enabling stakeholders to monitor dispute volumes, resolution cycle times, fraud detection rates, chargeback trends, and credit portfolio performance — with row-level security and drill-through capabilities.
- Strengthened governance using Microsoft Purview for automated cataloging, lineage tracking, sensitive data classification (PII, PCI, SOX), and policy enforcement — while configuring Entra ID RBAC, Key Vault for secrets management, and workspace-level security to protect regulated financial data.
- Monitored platform health using Azure Monitor, Fabric Monitoring Hub, Log Analytics, and Application Insights — configuring proactive alerting for pipeline SLA breaches, Spark failures, model drift signals, and storage thresholds to maintain 99.5%+ platform availability.
- Collaborated with Discover's data science, compliance, and business analytics teams in Agile delivery using JIRA, Confluence, and Azure DevOps — contributing to sprint planning, design reviews, code reviews, documentation, and production release coordination across cross-functional squads.

Environment: Microsoft Fabric (OneLake, Lakehouse, Warehouse, Data Pipelines, Dataflows Gen2, Eventstreams, Spark Notebooks, Power BI Semantic Models), Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Synapse Analytics, Azure Event Hubs, Azure OpenAI (GPT-4), Azure AI Search, Azure AI Foundry, Prompt Flow, Azure Machine Learning, MLflow, Azure Container Apps, AKS, Azure AI Content Safety, Microsoft Purview, Entra ID, Azure Key Vault, Azure Monitor, Log Analytics, Application Insights, Delta Lake, Parquet, Docker, GitHub Actions, Python, PySpark, Spark SQL, T-SQL, Power BI, JIRA, Confluence, Azure DevOps, Git, Agile/Scrum.

Client: LTI Mindtree – Microsoft Vendor **April 2020 - Aug 2023**
Role: Data Engineer **Mumbai, Hyderabad, India**

Responsibilities:

- Designed and developed 30+ scalable batch and near real-time data pipelines using Azure Data Factory with parameterized linked services, metadata-driven ingestion frameworks, and dynamic pipeline expressions to orchestrate end-to-end data movement and transformation across Azure Data Lake Storage Gen2, Azure Synapse Analytics, Azure SQL Database, and Snowflake — supporting analytics, reporting, and operational data needs across insurance and Xbox sales domains for Microsoft.
- Built enterprise-grade data ingestion frameworks for the insurance domain involving policy, claims, customer, billing, and agent-related datasets, extracting from multiple structured and semi-structured sources including SQL Server, MySQL, flat files, JSON, REST APIs, and Kafka topics — landing data into Bronze, Silver, and Gold zones within ADLS Gen2 following a medallion data lake architecture with Parquet and Delta file formats to ensure ACID compliance and schema evolution support.
- Engineered ETL/ELT pipelines to ingest and process high-volume Xbox sales transaction data across global regions, performing cleansing, standardization, deduplication, currency normalization, and business rule validations to build a single source of truth for revenue analytics, sales performance tracking, and operational reporting consumed by Microsoft business stakeholders — processing approximately 5+ TB of transactional data daily with a 99.5% pipeline success rate.
- Designed and implemented dimensional data models (star and snowflake schemas), curated reporting tables, slowly changing dimensions (SCD Type 1 and Type 2), data marts, and aggregated Gold-layer datasets within Azure Synapse Analytics dedicated SQL pools and Snowflake — enabling business teams to track claims processing efficiency, policy renewal rates, agent performance, Xbox sales velocity, and regional revenue distribution through downstream Power BI dashboards and ad-hoc SQL analysis.

- Developed event-driven and near real-time data processing solutions using Azure Event Hubs, Azure Stream Analytics, and Azure Service Bus to capture and process high-priority Xbox sales events and insurance claim alert triggers — enabling faster operational insights around sales spikes, promotional campaign impact, and claims anomaly detection, reducing reporting latency from hours to under 15 minutes for time-sensitive business decisions.
- Wrote complex Spark SQL, PySpark, and Python-based transformations on Azure HDInsight Spark clusters and Synapse Spark pools to process multi-terabyte datasets, performing window functions, time-series aggregations, complex joins across 20+ source tables, data enrichment, and business rule logic — improving data processing throughput by approximately 35% through partition pruning, broadcast joins, and caching optimization strategies.
- Implemented a comprehensive, multi-layer data quality framework across all pipeline stages — including source-to-target record count validation, schema conformance checks, referential integrity enforcement, duplicate detection, null threshold monitoring, and production load balancing — using Python scripts, Spark-based validation notebooks, and reusable SQL stored procedures that reduced data-related production incidents by approximately 40%.
- Created and executed unit tests, integration tests, and cross-system reconciliation scripts to validate row counts, business rule conformance, and end-to-end data consistency between source systems and target data warehouse layers — and developed Python automation to parse JSON payloads, generate daily control reports, and support pipeline capacity analysis and SLA tracking.
- Worked with Hadoop ecosystem tools including Hive for SQL-based querying on distributed datasets stored in HDFS, Sqoop for bidirectional bulk data movement between HDFS and relational databases (SQL Server, MySQL), HBase for low-latency key-value lookups on operational reference data, and Pig for data flow scripting — supporting big data processing, legacy platform integration, and migration workloads into Azure-native services.
- Orchestrated ADF pipelines using activities such as Lookup, GetMetadata, ForEach, If Condition, Execute Pipeline, Web Activity, and Stored Procedure to implement control flow patterns including incremental loads using watermark columns, full refresh patterns, conditional branching, and retry/failure notification logic — enabling self-healing pipeline behavior and reducing manual re-run interventions by approximately 50%.
- Built business-facing datasets, curated reporting feeds, and materialized views consumed by Power BI and Tableau dashboards for operational tracking, KPI monitoring, claims trend analysis, Xbox sales forecasting, and exploratory analysis — collaborating directly with business analysts, data consumers, and Microsoft reporting teams to translate complex business requirements into technical designs, user stories, and sprint-level implementation plans.
- Managed sensitive data handling and platform security using Azure Key Vault for secrets and connection string management, implemented row-level and column-level security in Synapse Analytics for PII protection, and configured role-based access controls across ADLS Gen2 containers and Synapse workspaces to comply with Microsoft's enterprise data governance and regulatory standards.
- Monitored pipeline execution health, job failures, SLA adherence, and compute resource utilization using Azure Monitor, Log Analytics, and ADF built-in monitoring dashboards — configuring alerts for pipeline latency breaches, Spark cluster memory pressure, and storage quota thresholds to enable proactive issue resolution and maintain 99.5%+ platform availability.
- Followed Git-based branching, pull request, and CI/CD deployment practices using Azure DevOps and Jenkins to maintain version-controlled, testable, and production-ready ADF pipelines, Spark notebooks, SQL scripts, and stored procedures — managing deployments across development, QA, staging, and production environments with ARM template parameterization and release gate approvals.

Environment: Azure Data Factory (ADF v2), Azure Data Lake Storage Gen2 (ADLS), Azure Blob Storage, Azure Synapse Analytics (Dedicated & Serverless SQL Pools, Synapse Spark), Azure HDInsight (Spark), Azure Stream Analytics, Azure Event Hubs, Azure Service Bus, Azure IoT Hub, Azure SQL Database, Azure Key Vault, Azure Monitor, Azure Log Analytics, Azure DevOps, Snowflake, Apache Kafka, Apache Spark (PySpark, Spark SQL), Hive, HBase, Sqoop, Pig, Hadoop (HDFS), SQL Server, MySQL, Python, Scala, SQL, T-SQL, Power BI, Tableau, JIRA, Confluence, Jenkins, Git, ARM Templates, Delta Lake, Parquet, Agile/Scrum.

Client: Bosch **May 2018 - March 2020**
Role: Data Engineer Intern **Bangalore, India.**

Responsibilities:

- Engineered end-to-end IoT telemetry ingestion pipelines for factory machine sensor data (temperature, vibration, operational readings) by developing Kafka producers and consumers in Python and Scala on EC2-hosted broker clusters within a private VPC, enabling real-time streaming of high-frequency industrial data into downstream processing and storage layers.
- Developed Spark Streaming applications in Scala deployed on multi-node AWS EMR clusters to consume raw telemetry events from Kafka topics, applying cleansing, enrichment, and aggregation transformations before persisting processed data into HBase for low-latency operational lookups and partitioned S3 data lake zones for durable cold storage and batch analytics.
- Implemented AWS Kinesis Data Streams as a parallel real-time ingestion channel for high-priority sensor alert events, with Lambda consumer functions performing threshold-based anomaly detection and triggering SNS notifications to factory operations teams when critical temperature or operational limits were breached — reducing incident response time from hours to near real-time.

- Built PySpark batch jobs on EMR with auto-scaling policies and spot instance strategies to process terabytes of historical IoT sensor datasets stored in S3, performing time-series aggregations, anomaly flagging, and statistical summaries to support machine health monitoring and predictive maintenance analytics for factory operations teams.
- Configured AWS Glue Crawlers to automatically discover and catalog raw and processed IoT sensor datasets across S3, maintaining a centralized Glue Data Catalog with partitioned table definitions that enabled factory analytics teams to run ad-hoc exploratory SQL queries via AWS Athena without provisioning dedicated compute resources.
- Designed and maintained a dimensional data model in AWS Redshift to store cleansed and aggregated machine performance metrics, optimizing sort and distribution keys for dashboard query patterns — enabling business stakeholders to track equipment efficiency, downtime patterns, and temperature threshold violations through operational BI dashboards.
- Deployed AWS Lambda functions triggered by S3 event notifications to perform automated data quality checks on newly ingested sensor files — validating record counts, schema conformance, and null thresholds — and writing validation results to an RDS PostgreSQL instance for audit trail tracking, reducing manual monitoring effort by approximately 60%.
- Maintained an AWS RDS (PostgreSQL) instance to store pipeline execution metadata, sensor-to-machine mapping reference tables, and data quality audit logs, enabling operational visibility into ingestion pipeline health and supporting end-to-end data lineage tracking across the platform.
- Monitoring CloudWatch dashboards and alarms to monitor EMR cluster utilization, Lambda invocation errors, Kinesis iterator age, and S3 storage growth — while enforcing least-privilege IAM policies and configuring VPC security groups and network ACLs to restrict data flow between ingestion, processing, and storage layers in alignment with Bosch's industrial data security standards.
- Collaborated with the IoT platform and analytics teams in an Agile delivery environment, contributing to sprint planning, technical design discussions, code reviews, and production deployments of the real-time data platform across development, staging, and production environments.

Environment: Apache Kafka, Apache Spark (PySpark, Spark Streaming, Spark SQL), AWS EMR, AWS S3, AWS Redshift, AWS Glue, AWS Athena, AWS Kinesis, AWS Lambda, AWS RDS (PostgreSQL), HBase, Amazon CloudWatch, AWS IAM, AWS VPC, EC2, SNS, Python, Scala, Java, SQL, VS Code, Git, Agile/Scrum.